LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Panel bucking, estimation of compressive loads for bucking a panel on simply supported structures
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Panel bucking, estimation of compressive loads for bucking a panel on simply supported structures. Rigidity due to end conditions on panel, study of heads, to bend panel with regards to variation in end condition shear buckling of plates
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of panel buckling under compressive load; Safety: Ensure load application is controlled; Supervision: Instructor monitors load limits and panel integrity
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary failure mode for thin skin panels under compression?
(b) Define simply supported boundary conditions for a structural panel.
(c) How does end rigidity affect the critical buckling load of a panel?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of panel bucking • Estimation of compressive loads • Simply supported structures analysis • Critical load calculation methods	L & Ds	PPT, IB	15 min
(b)	• Rigidity due to end conditions • Study of heads to bend panel • Variation in end condition effects • Shear buckling of plates	L & Ds	PPT, PDF	20 min
(c)	• Impact of boundary stiffness • Comparison of theoretical vs actual • Factors influencing buckling load • Role of shear in plate stability	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how end condition rigidity alters the critical buckling load.
(b) Describe the relationship between shear buckling and plate geometry.
(c) Identify two factors that increase the compressive load capacity of a panel.

12. Home Assignment and Follow Up	Prepare a brief note on the repair scheme considerations for a panel damaged due to buckling, referencing the next lesson plan (LP 17).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Assessment of damage and estimation of loads, preparation of repair scheme. Material specification, special tools required for carrying out repairs	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min